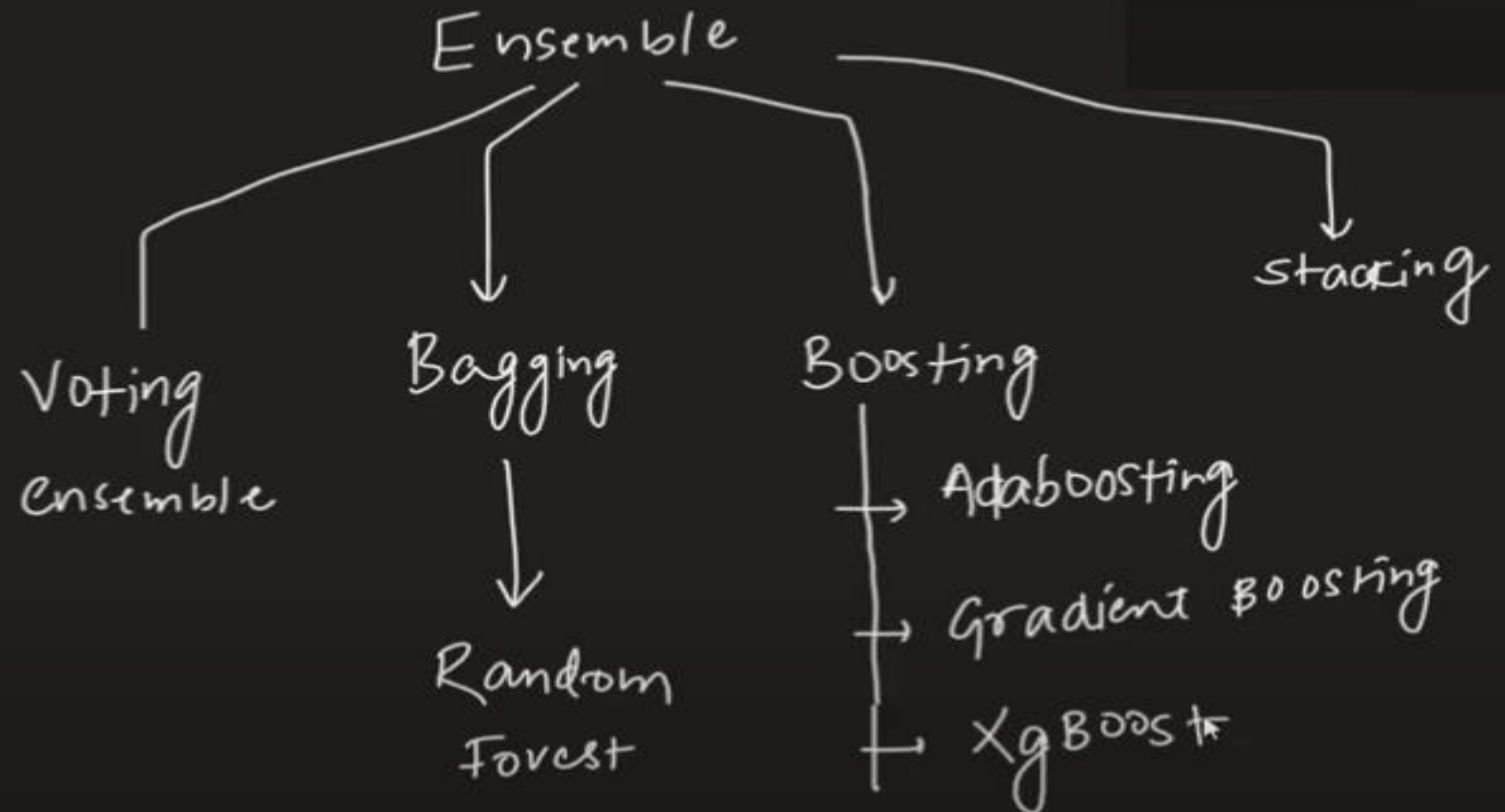


Ensemble Learning in Machine Learning

- Ensemble learning is a supervised learning technique used in machine learning to improve overall performance by combining the predictions from multiple models.
 - **Ensemble means group or collection**
 - A collection of machine learning models.
 - ❑ New model is generated from collection of ML models

Type of Ensemble Learning

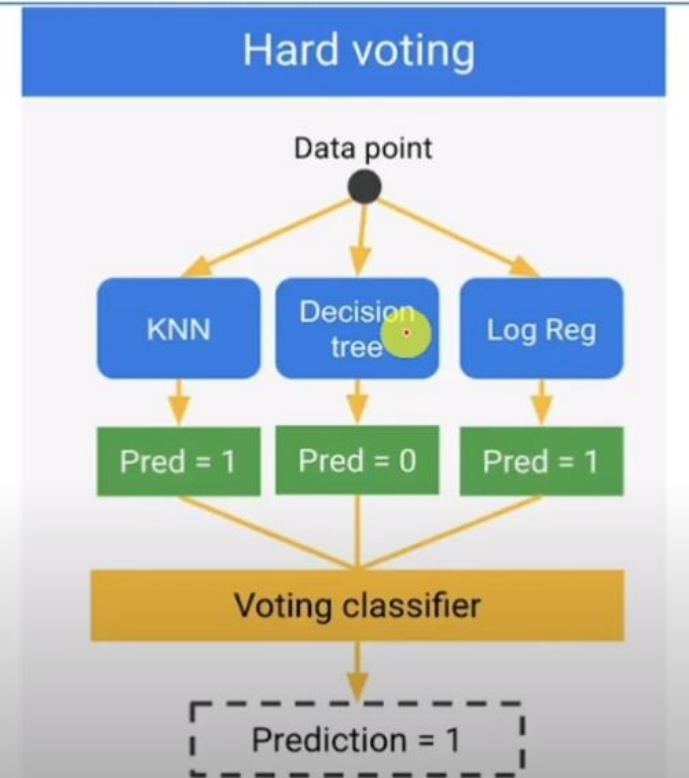


Ensemble Learning - Types of Ensemble Methods

- Voting (Averaging)
- Bootstrap aggregation (bagging)
- Random Forests
- Boosting
- Stacked Generalization (Blending)

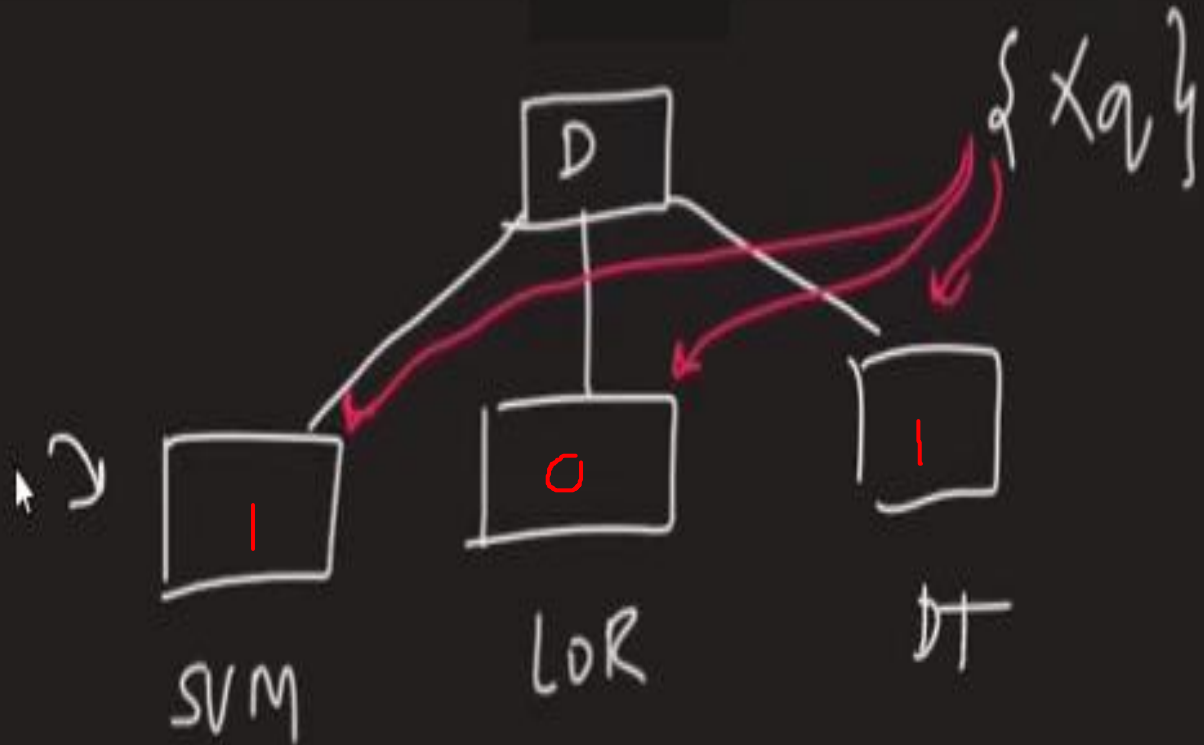
Ensemble Learning – Voting (Averaging)

- Voting is an ensemble machine learning algorithm that involves making a prediction that is the average (regression) or the sum (classification) of multiple machine learning models.



Voting

→ Base models
diff. algo

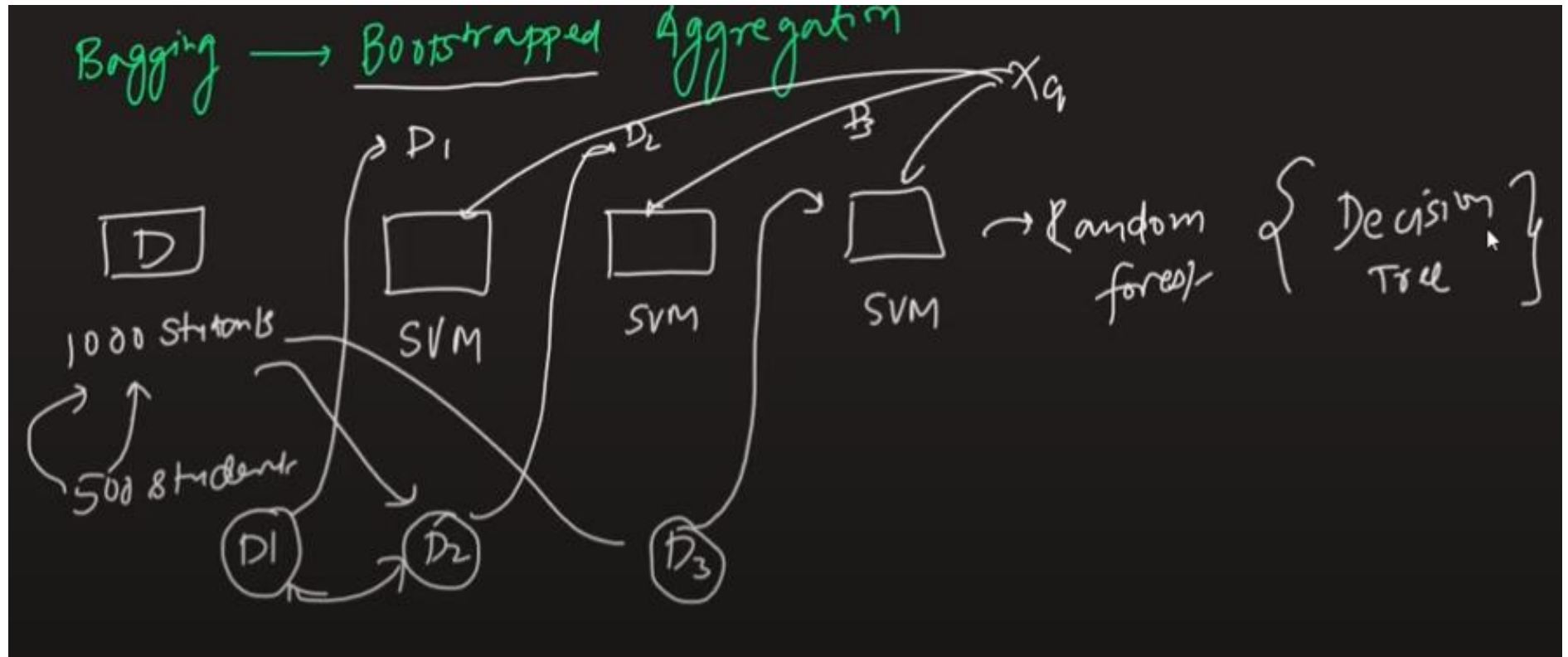


Bagging

Bagging is a method of ensemble modeling, which is primarily used to solve supervised machine learning problems. It is generally completed in two steps as follows:

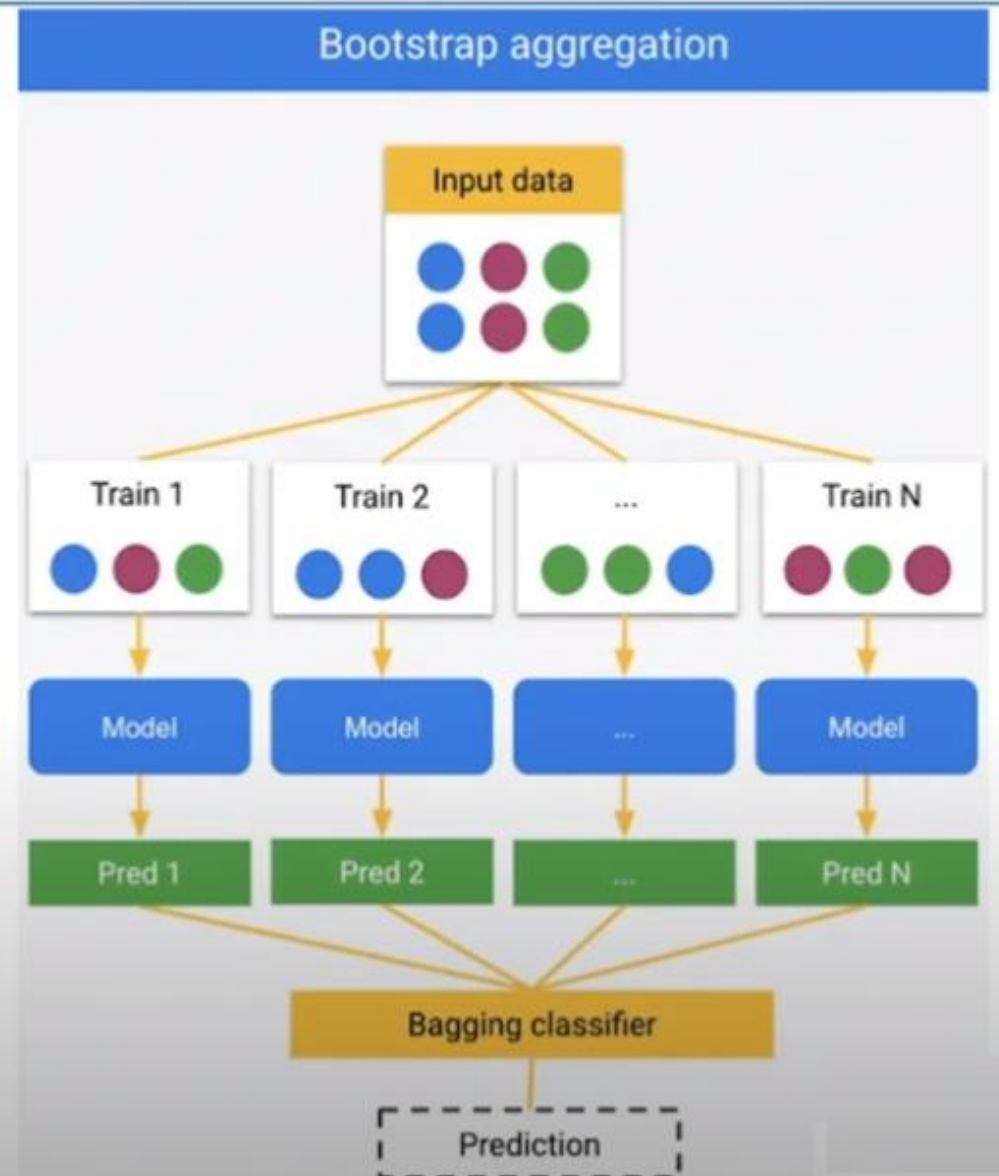
- **Bootstrapping:** It is a random sampling method that is used to derive samples from the data using the replacement procedure. In this method, first, random data samples are fed to the primary model, and then a base learning algorithm is run on the samples to complete the learning process.
- **Aggregation:** This is a step that involves the process of combining the output of all base models and, based on their output, predicting an aggregate result with greater accuracy and reduced variance.

Bagging



Ensemble Learning – Bootstrap aggregation (bagging)

- Bootstrap Aggregating, also known as bagging, is a machine learning ensemble meta-algorithm designed to improve the stability and accuracy of machine learning algorithms like classification and regression.
- It decreases the variance and helps to avoid overfitting.
- It is usually applied to decision tree methods.
- Bagging is a special case of the model averaging approach.



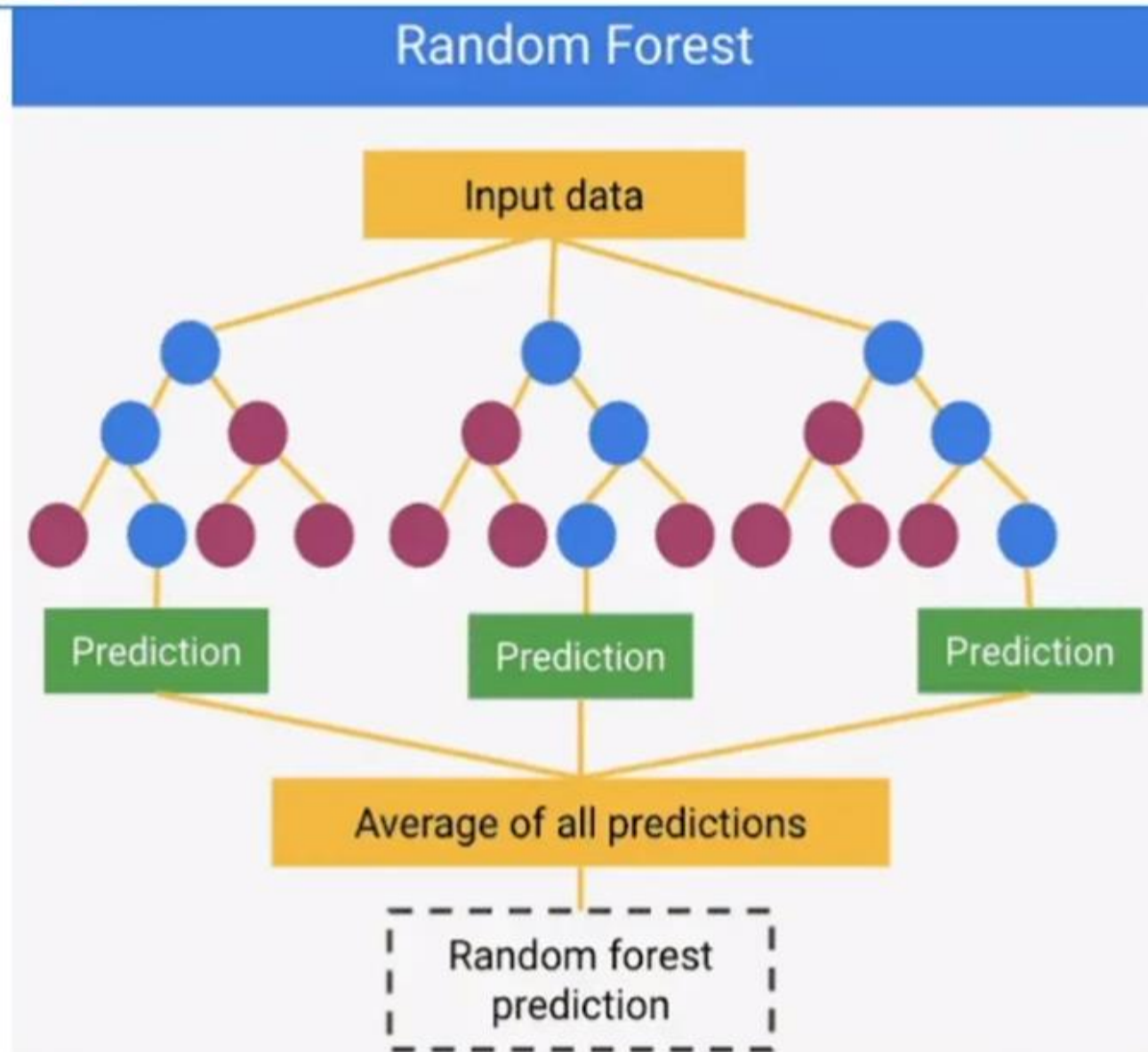
Bias, Variance, Overfitting, Underfitting

- **Bias** : Training errors
- **Variance** : Testing errors

- **Overfitting** :
 - ☐ Training errors : **Low**
 - ☐ Testing errors : **High**
- **Underfitting** :
 - ☐ Training errors : **High**
 - ☐ Testing errors : **Low**

Ensemble Learning – Random Forest

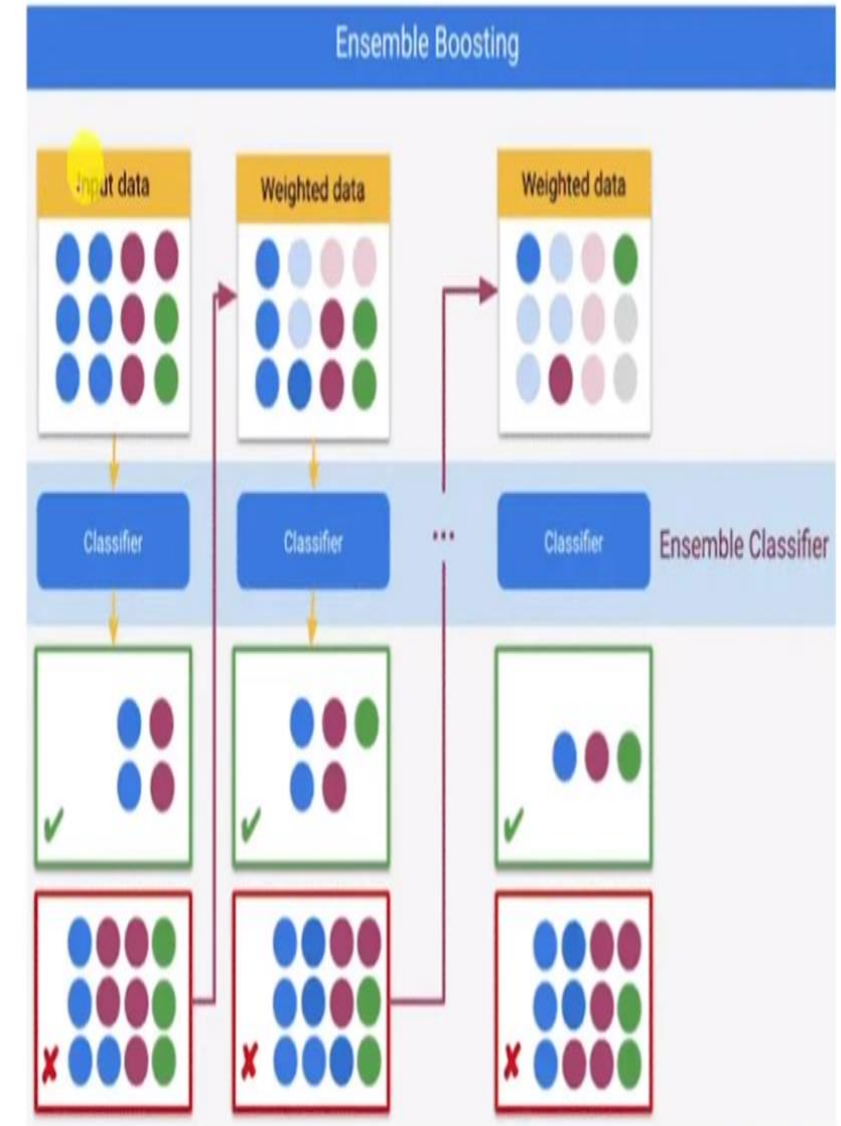
- Random forest is a commonly-used machine learning algorithm.
- A random forest is an ensemble learning method where multiple decision trees are constructed and then they are merged to get a more accurate prediction.



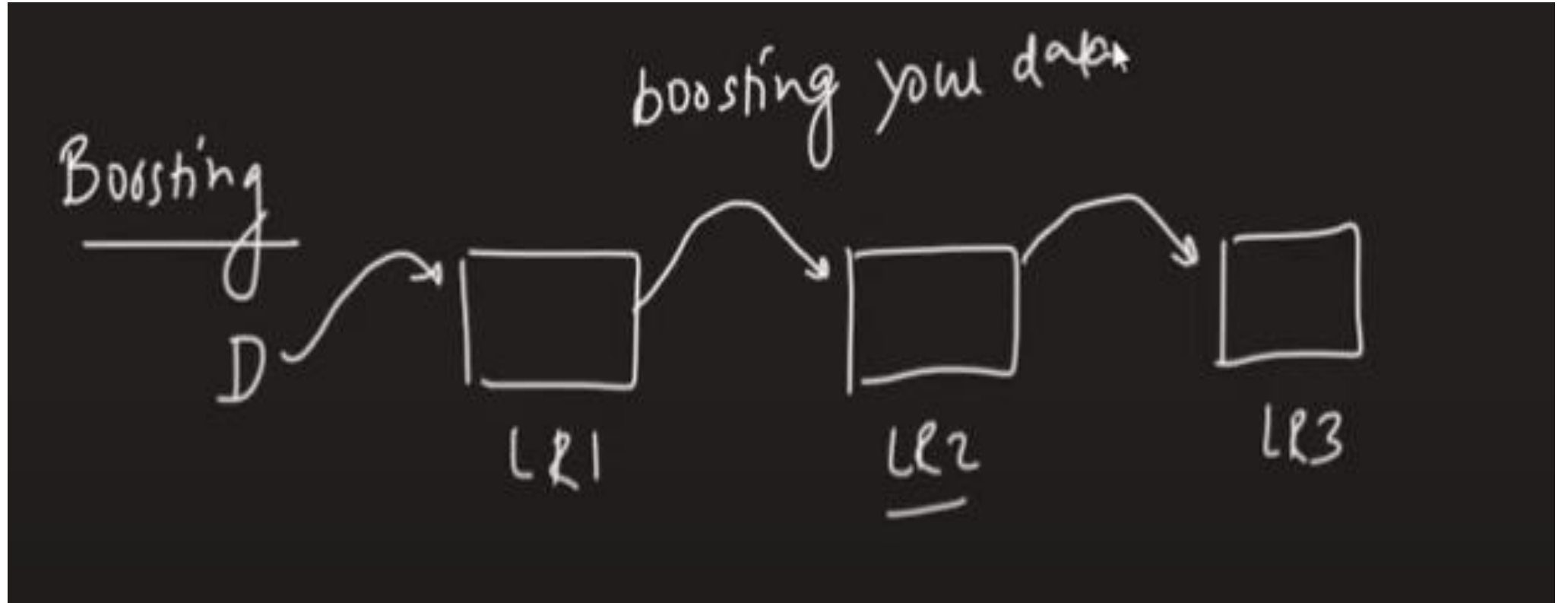
Ensemble Learning – Boosting

- Boosting is an ensemble modeling technique that attempts to build a strong classifier from the number of weak classifiers.
- It is done by building a model by using weak models in series.
- Firstly, a model is built from the training data.
- Then the second model is built which tries to correct the errors present in the first model.
- This procedure is continued and models are added until either the complete training data set is predicted correctly or the maximum number of models is added.

Ensemble Learning – Boosting



Boosting



Stacking in Ensemble Learning

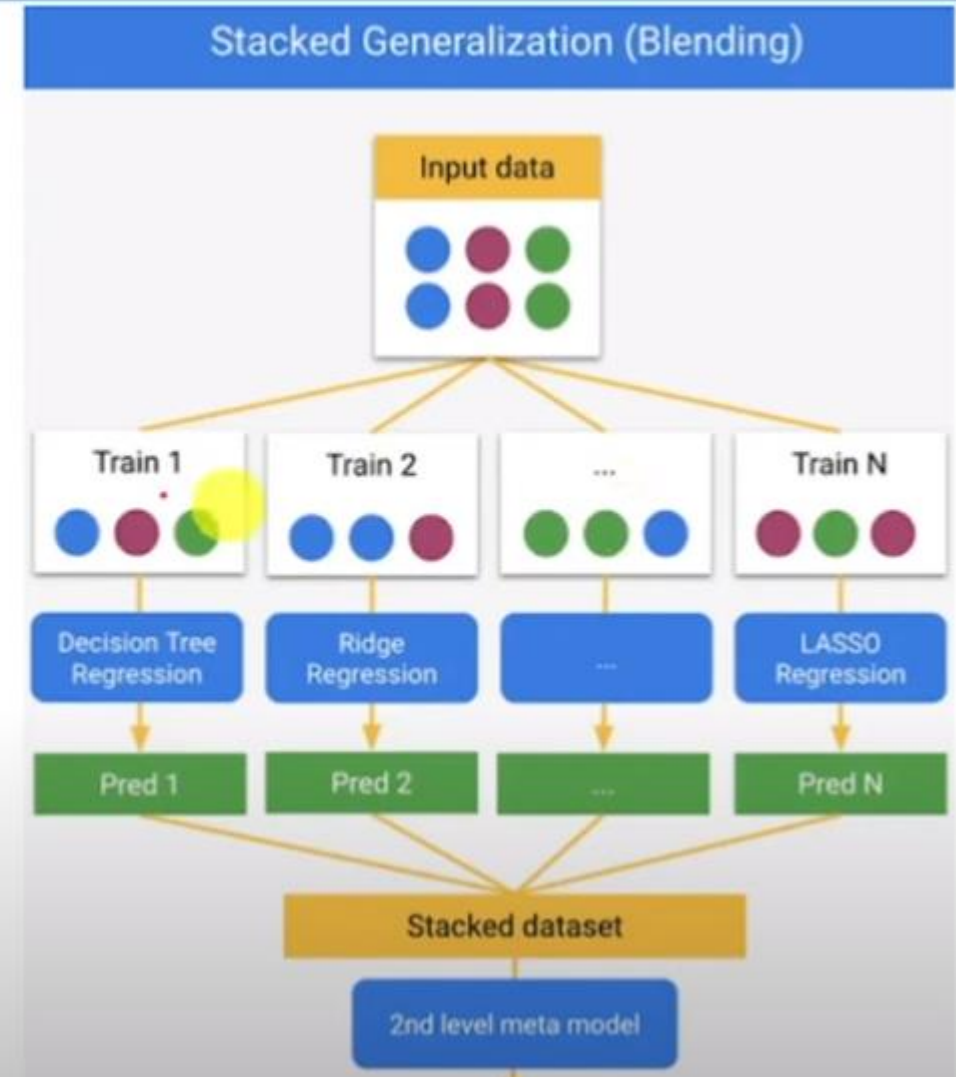
- Heterogeneous weak learners. E.g. Regression, SVM, KNN etc.
- Meta Model.
- Stacking enables us to train multiple models to solve similar problems, and based on their combined output, it builds a new model with improved performance.

Ensemble Learning – Stacked Generalization (Blending)

- Stacking, Blending and Stacked Generalization are all the same thing with different names. It is a kind of ensemble learning.
- In traditional ensemble learning, we have multiple classifiers trying to fit to a training set to approximate the target function.
- Since each classifier will have its own output, we will need to find a combining mechanism to combine the results.
- This can be through voting (majority wins), weighted voting (some classifier has more authority than the others), averaging the results, etc.

Ensemble Learning – Stacked Generalization (Blending)

- In stacking, the combining mechanism is that the output of the classifiers (Level 0 classifiers) will be used as training data for another classifier (Level 1 classifier) to approximate the same target function.
- Basically, you let the Level 1 classifier to figure out the combining mechanism.



Stacking

